

SIMATS ENGINEERING

CO5-ASSESSMENT TOOL3-Reverse Engineering

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO5: Design intelligent agents and real-time AI-based systems (chatbots, expert systems, emotion detection, edge AI, autonomous drones, and related applications) by selecting appropriate agent architectures, knowledge representation, and reasoning mechanisms to solve real-world problems. (BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Q1. The following is a conversation between a user and an AI-based flight-booking chatbot. Study the transcript given below and reverse-engineer the chatbot's design by: (a) writing its PEAS description, (b) classifying the type of agent it represents with justification, (c) identifying the likely component (NLU, Dialogue Manager, Knowledge Base, or NLG) responsible for each chatbot response, and (d) suggesting one enhancement using a learning-agent architecture.

Speaker	Message
User	I want to book a flight to Chennai next Friday.
Chatbot	Sure! What time would you like to depart, and do you have a preferred airline?
User	Morning flight, no preference.
Chatbot	I found 3 morning flights to Chennai on Friday. Shall I show you the cheapest option first?

Q2. The block diagram below shows the processing pipeline of a Real-Time Emotion Detection System. Reverse-engineer the system by: (a) explaining the function performed at each stage of the pipeline, (b) identifying the likely AI/ML technique used at each stage, (c) writing the PEAS description of the system, and (d) classifying the

Real-Time Emotion Detection System — processing pipeline



agent type (e.g., simple reflex, model-based) with justification.

Q3. An Expert System Shell for medical diagnosis contains the rule base, input facts, and reasoning trace given below. Study it and reverse-engineer the shell by: (a) identifying the type of chaining (forward/backward) used to reach the diagnosis, with justification, (b) identifying the expert-system-shell components illustrated (knowledge base, inference engine, explanation facility), (c) explaining how the shell would perform knowledge acquisition to add a new rule for dengue, and (d) explaining, with reasoning, whether backward chaining could reach the same diagnosis.

Rule	IF (condition)	THEN (conclusion)
R1	fever = yes AND cough = yes	suspect = flu
R2	suspect = flu AND body_ache = yes	diagnosis = Influenza
R3	fever = yes AND rash = yes	suspect = measles

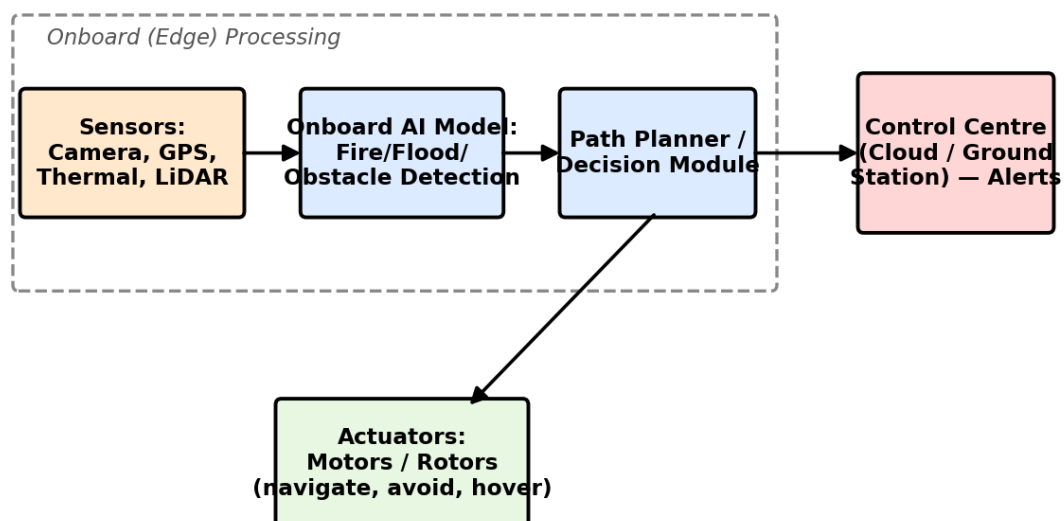
Given facts: fever = yes, cough = yes, body_ache = yes

Reasoning trace: R1 fires first (fever & cough matched) → suspect = flu. R2 then fires (suspect = flu & body_ache matched) → diagnosis = Influenza.

Explanation generated by the shell: “Diagnosis = Influenza because fever and cough indicated flu, and flu together with body ache confirmed Influenza.”

Q4. The diagram below shows the system architecture of an Autonomous Drone used for Disaster Detection with Edge AI deployment. Reverse-engineer the design by: (a) writing the PEAS description of the drone agent, (b) classifying the agent type (goal-based/utility-based) with justification, (c) identifying which components run on the edge versus the cloud and justifying why on-edge processing was chosen for detection and obstacle avoidance, and (d) identifying whether the agent's control architecture is reactive, deliberative, or hybrid, with justification.

Autonomous Drone — Disaster Detection (Edge AI) system architecture



Q5. The table below shows sample output logs from an AI-based Deepfake & Face Authenticity Verification system across five video frames. Study the output and reverse-engineer the system by: (a) identifying the likely input features used by the model (e.g., blink rate, texture/artifact score), (b) identifying the type of model/technique likely used to produce these outputs, (c) explaining how the final Real/Fake decision may have been derived from the frame-level confidence scores,

and (d) suggesting one way a swarm/ensemble-based approach could improve the system's reliability.

Frame No.	Blink Rate (per 10s)	Texture/Artifact Score	Fake Confidence (%)	Prediction
1	4	0.12	8	Real
2	1	0.61	74	Fake
3	0	0.83	91	Fake
4	5	0.09	5	Real
5	2	0.55	68	Fake

Overall system verdict: Fake (3 of 5 frames flagged — majority vote)

Code of Conduct:

*I **G DHANUSH (192512132)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G DHANUSH

(192512132)

RUBRICS FOR CALCULATION

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Max Marks
Identification of Agent Type & PEAS (correctly inferred from the given artifact)	Correctly identifies the agent type and gives a complete, accurate PEAS description, fully consistent with the given artifact.	Identifies the agent type and PEAS description correctly with only minor omissions.	Partially correct identification of agent type/PEAS; some elements are missing or inconsistent with the artifact.	Agent type/PEAS identification is largely incorrect or not attempted.	10
Identification of Architecture & Components (mapping artifact evidence to system components)	Accurately identifies all relevant architectural components and correctly maps each to specific evidence in the artifact.	Identifies most components correctly with reasonable mapping to the artifact; minor gaps present.	Identifies some components; mapping to the artifact is weak or partially incorrect.	Little or no correct identification of architecture/components.	10
Identification of Underlying Technique/Algorithm	Correctly reverse-engineers the	Correctly identifies the	Identification is only partially	Technique/mechanism is incorrectly	10

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
Algorithm/Reasoning Mechanism	technique, algorithm, or reasoning mechanism (e.g., chaining type, ML technique) with clear supporting evidence.	technique/mechanism with adequate, mostly accurate justification.	correct or lacks sufficient supporting evidence.	identified or not addressed.	
Quality of Justification & Use of Evidence (linking conclusions back to the given artifact)	Every conclusion is clearly justified with specific, accurate references to the given artifact.	Most conclusions are justified with reasonable references to the artifact.	Justification is vague or only loosely connected to the artifact.	Conclusions are stated with little or no justification or evidence.	10
Clarity & Completeness of Presentation (addresses all sub-parts of the question in an organised manner)	All sub-parts are addressed completely, in a clear, well-organised, and logically structured manner.	Most sub-parts are addressed clearly with minor gaps in organisation or completeness.	Some sub-parts are missing or the presentation is unclear/poorly organised.	Response is incomplete, disorganised, or largely missing.	10
				Total	50

1. Reverse Engineering of AI-Based Flight-Booking Chatbot

The chatbot receives the user's request to book a flight to Chennai on the following Friday. It asks for additional preferences and then searches for suitable morning flights.

(a) PEAS Description

Component	Description
P – Performance Measure	Accurate flight recommendations, successful booking, minimum response time, user satisfaction, and suitable price selection.
E – Environment	User, airline databases, flight schedules, booking system, and available flights.
A – Actuators	Text responses, displaying flight options, recommending flights, and initiating booking actions.
S – Sensors	User messages, destination, travel date, departure-time preference, airline preference, and flight availability data.

(b) Type of Agent

The chatbot can be classified as a **Goal-Based Agent**.

Reason :

The chatbot works toward achieving the user's goal of booking a suitable flight. It collects information such as the preferred departure time and airline before identifying available flights. The conversation shows that after receiving the user's preference for a morning flight, it finds three suitable flights and offers the cheapest option first. Therefore, its actions are directed toward achieving a specific goal.

(c) Components Responsible for the Responses

Chatbot Response/Function	Likely Component	Reason
Understanding "book a flight to Chennai next Friday"	NLU	Extracts destination and travel date from the user's message.
Asking for departure time and preferred airline	Dialogue Manager	Determines what additional information is required to continue the conversation.
Finding three morning flights	Knowledge Base	Stores or accesses flight schedules and availability information.
"Shall I show you the cheapest option first?"	NLG	Generates a natural-language response for the user.

(d) Enhancement Using a Learning-Agent Architecture

A **learning component** can be added to record the user's previous preferences, such as frequently selected airlines, preferred departure times, or price ranges.

For example, if the user repeatedly chooses the cheapest morning flight, the system can learn this preference and automatically prioritize similar options in future searches.

Result: The chatbot becomes more personalized and improves its recommendations over time.

2. Reverse Engineering of a Real-Time Emotion Detection System

(a) Function Performed at Each Stage

Stage	Function
Input Acquisition	Captures real-time input such as an image or video frame.
Pre-processing	Improves the input by resizing, normalization, noise reduction, or face detection.
Feature Extraction	Identifies important emotional features such as facial expressions, eyes, eyebrows, and mouth position.
Emotion Classification	Classifies the detected emotion, such as happy, sad, angry, surprised, or neutral.
Output/Decision	Displays or sends the final detected emotion.

(b) Likely AI/ML Technique

1. **Face Detection:** Computer vision techniques or a deep-learning face detector.
2. **Pre-processing:** Image-processing techniques such as normalization and resizing.
3. **Feature Extraction:** A **Convolutional Neural Network (CNN)** can extract important facial features.
4. **Emotion Classification:** A CNN or other supervised machine-learning classifier can classify the emotion.
5. **Output:** The predicted emotion with its confidence score is presented to the user or another system.

(c) PEAS Description

Component	Description
P – Performance Measure	High emotion-detection accuracy, low delay, and reliable real-time performance.
E – Environment	Camera, users, lighting conditions, facial expressions, and surrounding environment.

Component	Description
A – Actuators	Displaying the detected emotion, generating an alert, or sending the result to another system.
S – Sensors	Camera or video input capturing facial images and expressions.

(d) Agent Type

The system can be classified as a **Model-Based Agent**.

Reason :

The system processes the current camera input using an internal AI/ML model that has learned patterns associated with different emotions. It does not simply use a direct condition-action rule. Instead, the model interprets the observed facial features and predicts the most appropriate emotional state.

3. Reverse Engineering of the Medical Expert System Shell

- I. The given rules and reasoning trace are:
- II. R1: Fever + cough $\rightarrow$ suspect flu
- III. R2: Suspect flu + body ache $\rightarrow$ diagnosis Influenza
- IV. Given facts: fever, cough, and body ache
- V. R1 fires first, followed by R2, resulting in Influenza.

(a) Type of Chaining Used

The system uses **Forward Chaining**.

Justification:

The reasoning begins with the known facts:

fever = yes

cough = yes

body_ache = yes

The inference engine checks these facts against the rules.

Step 1:

R1:

IF fever = yes AND cough = yes THEN suspect = flu

Since both conditions are true:

suspect = flu

is inferred.

Step 2:

R2:

IF suspect = flu AND body_ache = yes THEN diagnosis = Influenza

Both conditions are now satisfied.

Therefore:

diagnosis = Influenza

Thus, the system starts with facts and moves forward toward the conclusion, which is the characteristic of **forward chaining**.

(b) Expert System Shell Components

1. Knowledge Base

The knowledge base contains:

- Rules R1, R2, and R3.
- Facts such as fever = yes, cough = yes, and body_ache = yes.
- The rules and facts represent the domain knowledge used for diagnosis.

2. Inference Engine

The inference engine applies the rules to the available facts.

It first activates R1 and derives:

suspect = flu

Then it activates R2 and derives:

diagnosis = Influenza

3. Explanation Facility

The explanation facility provides the reason for the final diagnosis.

The system explains that fever and cough indicate flu, and flu together with body ache confirms Influenza.

CO5_AT3_ReverseEngineering

(c) Knowledge Acquisition for Adding a New Dengue Rule

To add a new rule, the expert-system shell would obtain knowledge from medical experts and identify the symptoms and conclusion associated with dengue.

For example:

- i. **IF fever = yes AND rash = yes AND body_ache = yes THEN diagnosis = dengue**

- ii. The knowledge acquisition process would be:
- iii. Consult a medical expert.
- iv. Identify the relevant symptoms and conditions.
- v. Convert the medical knowledge into an IF–THEN rule.

Add the rule to the knowledge base.

- vi. Test the rule using suitable facts.
- vii. Validate the result with the expert.
- viii. Thus, the knowledge base can be expanded without redesigning the entire expert system.

(d) Can Backward Chaining Reach the Same Diagnosis?

Yes, backward chaining can also reach the diagnosis of Influenza.

It would start with the goal:

Goal: diagnosis = Influenza

Then it searches for a rule that can prove this goal.

From R2:

suspect = flu AND body_ache = yes → diagnosis = Influenza

The system now needs to prove:

- ✓ suspect = flu
- ✓ body_ache = yes
- ✓ Body ache is already given as a fact.
- ✓ To prove **suspect = flu**, it uses R1:
- ✓ **fever = yes AND cough = yes → suspect = flu**
- ✓ Both fever and cough are available as facts.

Therefore, backward chaining can also prove:

diagnosis = Influenza

4. Reverse Engineering of Autonomous Drone for Disaster Detection

(a) PEAS Description

Component	Description
P – Performance Measure	Accurate disaster detection, fast response, successful navigation, obstacle avoidance, safety, and efficient energy usage.
E – Environment	Disaster areas, buildings, roads, obstacles, weather conditions, victims, and communication networks.
A – Actuators	Propellers, motors, navigation controls, camera controls, and communication modules.
S – Sensors	Camera, GPS, obstacle sensors, altitude sensors, temperature sensors, and other environmental sensors.

(b) Agent Type

The drone can be classified as a **Utility-Based Agent**.

Justification:

The drone may have several possible actions, such as choosing different routes, avoiding obstacles, continuing surveillance, or transmitting data. It must select the action that provides the best overall result by considering factors such as safety, speed, energy consumption, and detection accuracy.

Therefore, a utility-based agent is suitable because it can evaluate and select the most beneficial action.

(c) Edge Processing vs Cloud Processing

Components Running on the Edge

The following operations can run directly on the drone:

- Camera data processing
- Disaster detection

- Object detection
- Obstacle detection
- Obstacle avoidance
- Immediate navigation decisions

Components Running in the Cloud

- ✓ The cloud can be used for
 - ✓ Long-term data storage
 - ✓ Model training and updates
-
- Large-scale analysis
 - Historical disaster analysis
 - Monitoring data from multiple drones

Why Edge Processing Is Used

Edge AI is used because disaster detection and obstacle avoidance require **immediate decisions**.

Sending every image to the cloud may introduce communication delays. During an emergency, a delayed decision could affect the drone's safety and performance.

- ✓ Therefore, on-edge processing provides:
- ✓ Lower latency
- ✓ Faster response
- ✓ Reduced dependence on internet connectivity
- ✓ Improved real-time obstacle avoidance

(d) Control Architecture

The drone most likely uses a **Hybrid Architecture**.

Reactive Part:

Immediately responds to obstacles and sensor readings.

Example:

Obstacle detected → immediately change direction

Deliberative Part:

Plans routes and determines how to achieve the larger disaster-monitoring objective.

Therefore, the drone combines immediate reactions with goal-oriented planning, making a **hybrid architecture** appropriate.

5. Reverse Engineering of Deepfake and Face Authenticity Verification System

(a) Likely Input Features

The model likely uses the following features:

1. Blink Rate

Blink rate measures how frequently the person's eyes blink.

From the table:

- Frame 1: Blink rate = 4 → Real
- Frame 3: Blink rate = 0 → Fake
- An unusual blink pattern may contribute to a higher probability of a manipulated face.

2. Texture/Artifact Score

This feature measures visual irregularities or artifacts in the face.

Examples may include:

- Unnatural skin texture
- Blurred facial regions
- Inconsistent facial boundaries
- Visual artifacts
- Frames with higher artifact scores have higher fake confidence. For example, Frame 3 has an artifact score of 0.83 and fake confidence of 91%, resulting in a Fake prediction.

3. Additional Possible Features

The model may also analyze:

- Facial movement consistency
- Eye movement
- Lip synchronization
- Facial texture patterns

- Frame-to-frame consistency

(b) Likely Model or Technique

- ✓ A likely technique is a **Deep Learning-based Computer Vision Model**, such as a **Convolutional Neural Network (CNN)**.
- ✓ The system processes facial frames and extracts visual features such as texture and artifacts. These features are then used to predict whether each frame is Real or Fake.
- ✓ The blink rate can also be combined with image-based features to improve the classification.
- ✓ Therefore, the system may use a **CNN-based classifier with multiple input features**.
- ✓

(c) How the Final Real/Fake Decision Is Derived

The system first performs classification on every frame individually.

Frame	Prediction
Frame 1	Real
Frame 2	Fake
Frame 3	Fake
Frame 4	Real
Frame 5 Fake	

The final decision is obtained using **Majority Voting**.

Since most frames are classified as Fake:

Final Verdict=**Fake**

This matches the overall system verdict given in the assignment.

(d) Improvement Using Swarm/Ensemble-Based Approach

The reliability can be improved using an **ensemble approach**.

Instead of using only one model, multiple AI models can independently analyze the same video.

For example:

Model 1: Detects texture artifacts.

Model 2: Analyzes blink and eye movement.

Model 3: Checks facial movement consistency.

Model 4: Analyzes frame-to-frame changes.

Each model produces a prediction.

The final result can be calculated using:

Majority voting, or

Weighted voting based on model confidence.

For example, if three out of four models classify the video as Fake, the system can give a stronger Fake prediction.

Conclusion

An ensemble or swarm-based approach can improve reliability because the final decision does not depend completely on a single model or feature. Different models can detect different types of deepfake artifacts, leading to a more robust verification system.